Data-Backed Journal-Style Replication of *Collaboration Networks, Structural Holes, and Innovation*

Replication Paper Project

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Abstract

This paper presents a local calculated replication of Collaboration Networks, Structural Holes, and Innovation: A Longitudinal Study using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 2,378 observations across 177 entities, and the batch headline estimate is -0.0125 for Direct Ties with p = ¡ .001. The local citation-network analogue reverses the original alliance-network pattern in important ways: direct ties are negative, structural holes are positive, and indirect ties are weak once stock and scale controls are imposed. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: interfirm networks; direct ties; indirect ties; structural holes; innovation; local calculated analogue

1 Introduction

This replication begins from a paper-first reading of Collaboration Networks, Structural Holes, and Innovation: A Longitudinal Study and treats the source article as an empirical argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks how direct ties, indirect ties, and structural holes in an interorganizational network shape subsequent innovation

(Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The study is worth revisiting because The replication matters because network arguments are highly sensitive to what counts as a tie. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 2,378 observations across 177 entities and spans 1991 to 2014 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Ahuja, 2000a,b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent network translation that makes the tie-definition shift explicit and shows how the innovation coefficients change when knowledge links replace collaborative alliances. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Ahuja, 2000a,b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper belongs to a top-journal literature on interorganizational networks, structural holes, and innovative performance. (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because the current rerun asks whether Ahuja's network logic survives when the interfirm relation is reconstructed from processed patent knowledge links rather than from recorded collaborative alliances. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than

on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). Because this paper is network-theoretic at its core, the review adds top-journal studies on alliances, network position, and interfirm knowledge diffusion so the citation-link translation is interpreted against the right conceptual backdrop. (Ahuja, 2000b; Ahuja and Katila, 2001). A second paper-specific literature block is added so that the manuscript's topic-matched references are carried into the final bibliography explicitly rather than being silently implied by the broader shared review. (Stuart and Podolny, 1996; Rosenkopf and Almeida, 2003). A second paper-specific literature block is added so that the manuscript's topic-matched references are carried into the final bibliography explicitly rather than being silently implied by the broader shared review. (Schilling and Phelps, 2007; Almeida and Kogut, 1999). A second paper-specific literature block is added so that the manuscript's topic-matched references are carried into the final bibliography explicitly rather than being silently implied by the broader shared review. (Rivkin and Siggelkow, 2003; Lavie et al., 2011).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Ahuja, 2000a). The original study can be summarized as a mechanism paper with a sharp empirical core. The original article studies direct ties, indirect ties, and structural holes in an interfirm network. Subsequent innovation is the central outcome. The design is longitudinal and network-based. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Ahuja, 2000a; March, 1991; Levitt

and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Direct ties, indirect ties, and structural holes are the core constructs. The local reconstruction therefore adopts the following bounded decision: The local rerun reconstructs the network from processed patent knowledge links. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Innovation output is the central dependent variable. The local reconstruction therefore adopts the following bounded decision: Next-year patenting is used as the executable innovation outcome. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The source network is collaborative. The local reconstruction therefore adopts the following bounded decision: The paper treats the tie-definition shift as a visible translation rather than hiding it. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly

where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

DirectSourceFact	LocalReplicationDecision
Direct ties, indirect ties, and structural holes are the core constructs.	The local rerun reconstructs the network from processed patent knowledge links.
Innovation output is the central dependent variable.	Next-year patenting is used as the executable innovation outcome.
The source network is collaborative.	The paper treats the tie-definition shift as a visible translation rather than hiding it.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Rolling firm-year network reconstructed from processed focal patents and backward patent links. Current-year patent and publication counts plus financial controls from the local public-firm panel. Next-year patent output used as the executable innovation outcome (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 2,378 observations across 177 entities and covers 1991 to 2014 in the local executable panel. Every numeric claim in the results section is intended to trace back to one of those verified evidence layers rather than to a hand-copied number from the article (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny,

1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local analogue reconstructs a rolling firm-to-firm knowledge-link network from backward patent ties among public firms and then maps direct ties, indirect ties, and structural holes into a firm-year innovation panel. The local sample frame is Public-firm year panel with rolling knowledge-link network measures and next-year patent output The observed time window in the executable panel is 1991 to 2014 in the local executable panel The outcome is Next-year patent count at the firm-year level (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Ahuja, 2000a; Hall et al., 2005).

3.1 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Ahuja (2000). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005). The analysis panel used in this paper serves a specific empirical role: it is needed to estimate the local model and regenerate descriptive statistics. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The coefficient export used in this paper serves a specific empirical role: it is needed to recover the reported local coefficients and p-values. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample

moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The sample summary used in this paper serves a specific empirical role: it is needed to audit sample construction and raw grouped moments. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The fit summary used in this paper serves a specific empirical role: it is needed to check model diagnostics when a fit file is available. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

Stage	NeededFor	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	Yes	Executable local result file

Stage	NeededFor	Exists	Status
Sample summary	Audit sample construction and raw grouped moments	Yes	Executable local result file
Fit summary	Check model diagnostics when a fit file is available	Yes	Executable local result file

Verified evidence layers used to support the paper's calculations.

3.2 Sample Reconstruction

The local analysis file contains 2,378 observations drawn from 177 entities. The observed time span runs from 1991 to 2014. Selected sample moments are Direct Ties = 35.89, Indirect Ties = 52.56, Structural Holes = 0.256, Next Patents = 228.2 (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	2,378
Unique Firms	177
Start Year	1991
End Year	2014
Mean Direct Ties	35.89
Mean Indirect Ties	52.56
Mean Structural Holes	0.256
Mean Next Patents	228

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Direct ties is operationalized locally as Distinct knowledge-link neighbors and appears in the executable dataset as direct_ties. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Count unique neighbors in the rolling citation-link network.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Indirect ties is operationalized locally as Reachable second-order neighbors and appears in the executable dataset as indirect_ties. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Count unique second-order neighbors excluding direct ties.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Structural holes is operationalized locally as One minus neighbor density and appears in the executable dataset as structural_holes. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Compute one minus the density among each focal firm's neighbors.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In

replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Innovation output is operationalized locally as Next-year patent count and appears in the executable dataset as next_patents. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Lead current patent counts by one year within each firm.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Firm-year controls is operationalized locally as Patent stock, assets, and R&D intensity and appears in the executable dataset as log_pat_stock; log_assets; rd_intensity. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Reuse the local firm-year stock and financial controls.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Direct ties	Distinct knowledge-link neighbors	Count unique neighbors in the rolling citation-link network.
Indirect ties	Reachable second-order neighbors	Count unique second-order neighbors excluding direct ties.
Structural holes	One minus neighbor density	Compute one minus the density among each focal firm's neighbors.
Innovation output	Next-year patent count	Lead current patent counts by one year within each firm.
Firm-year controls	Patent stock, assets, and R&D intensity	Reuse the local firm-year stock and financial controls.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Ahuja (2000). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Direct ties, the manuscript uses Distinct knowledge-link neighbors under the exact local implementation direct_ties. The derivation step is: Count unique neighbors in the rolling citation-link network.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Indirect ties, the manuscript uses Reachable second-order neighbors under the exact local implementation indirect_ties. The derivation step is: Count unique second-order neighbors excluding direct ties.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Structural holes, the manuscript uses One minus neighbor density under the exact local implementation structural_holes. The derivation step is: Compute one minus the density among each focal firm's neighbors.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Innovation output,

the manuscript uses Next-year patent count under the exact local implementation next_patents. The derivation step is: Lead current patent counts by one year within each firm.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Firm-year controls, the manuscript uses Patent stock, assets, and R&D intensity under the exact local implementation log_pat_stock; log_assets; rd_intensity. The derivation step is: Reuse the local firm-year stock and financial controls.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The current dataset yields a richer descriptive picture than a single mean or coefficient can convey. Next-year patents has mean 228.2, standard deviation 519.3, and an interquartile range from 20.00 to 226.4; Direct ties has mean 35.89, standard deviation 21.87, and an interquartile range from 18.00 to 51.00; Indirect ties has mean 52.56, standard deviation 26.60, and an interquartile range from 31.00 to 75.00; Structural holes has mean 0.256, standard deviation 0.156, and an interquartile range from 0.158 to 0.354 (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time-series view is equally informative. FirmYears changes from 132.0 in the first observed period to 75.00 in the last observed period; MeanDirectTies changes from 23.24 in the first observed period to 35.76 in the last observed period; MeanIndirectTies changes from 68.59 in the first observed period to 27.33 in the last observed period . These temporal shifts help explain why a paper-comparable replication has to describe the support of the data before turning to regression output (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The regressor-quantile analysis shows that mean outcomes range from 12.67 to 657.9 across the support of the focal regressor, while the mean regressor itself ranges from 5.306 to 66.92. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (Ahuja, 2000a; Jaffe, 1986; Hall et al.,

2005; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation structure is reported in a compact form because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. Next-year patents has missing share 0.00%; Direct ties has missing share 0.00%; Indirect ties has missing share 0.00% . At the other end of the support, Next-year patents remains close to complete with missing share 0.00%; Direct ties remains close to complete with missing share 0.00%; Indirect ties remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather than assumed. TEXAS INSTRUMENTS INC contributes 24 observations; TEXTRON INC contributes 24 observations; ABBOTT LABORATORIES contributes 24 observations; RTX CORP contributes 24 observations; WHIRLPOOL CORP contributes 24 observations . The purpose of reporting these concentrations is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005). The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 5.306 and the mean outcome is 12.67 across 500 observations. In the highest observed quantile, the corresponding values are 66.92, 657.9, and 451 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because network arguments are highly sensitive to what counts as a tie. If the local panel were badly fragmented, dominated by miss-

ingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005).

Variable	Missing	MissingShare
Next-year patents	0	0.00%
Direct ties	0	0.00%
Indirect ties	0	0.00%
Structural holes	0	0.00%
Log patent stock	0	0.00%
Log assets	0	0.00%
R&D intensity	0	0.00%
Current-year patents	0	0.00%
Current-year publications	0	0.00%

Missingness profile for variables analyzed in the body of the paper.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Next-year patents	2,378	228.2	519.3	20.00	78.50	226.4	0	7662
Direct ties	2,378	35.89	21.87	18.00	37.00	51.00	0	96
Indirect ties	2,378	52.56	26.60	31.00	54.00	75.00	0	105
Structural holes	2,378	0.256	0.156	0.158	0.255	0.354	0	1
Log patent stock	2,378	5.803	2.097	4.833	6.254	7.236	0	10
Log assets	2,378	9.388	1.341	8.480	9.310	10.26	4	14
R&D intensity	2,378	0.0417	0.0403	0.0134	0.0281	0.0594	0	1
Current-year patents	2,378	218.2	489.0	20.00	77.00	217.1	0	7661
Current-year publications	2,378	146.6	301.6	3.500	23.00	154.0	0	2459

Main descriptive statistics discussed in the body of the paper.

Period	FirmYears	MeanDirectTies	MeanIndirectTies	MeanStructuralHoles	MeanNextPatents
1991	132	23.24	68.59	0.377	118

Period	FirmYears	MeanDirectTies	MeanIndirectTies	MeanStructuralHoles	MeanNextPatents
1992	130	28.17	68.15	0.352	122
1993	126	31.56	69.62	0.321	130
1994	123	34.60	66.11	0.292	129
1995	122	37.31	69.41	0.287	140
1996	123	37.32	72.59	0.287	130
1997	120	38.48	71.18	0.283	170
1998	113	38.71	66.27	0.282	183
1999	106	38.83	57.91	0.270	195
2000	102	35.86	47.63	0.234	209
2001	100	38.10	48.88	0.233	219
2002	98	38.04	53.18	0.238	226
2003	100	39.50	51.58	0.235	216
2004	98	39.08	49.20	0.241	195
2005	93	40.26	45.68	0.235	250
2006	89	40.40	39.73	0.229	232
2007	83	40.12	33.93	0.205	258
2008	78	37.77	31.51	0.198	292
2009	75	35.28	30.99	0.186	387
2010	75	35.07	30.45	0.181	374
2011	72	34.72	30.69	0.179	425
2012	70	34.91	25.17	0.179	463
2013	75	35.12	26.45	0.175	468
2014	75	35.76	27.33	0.178	463

Observed time patterns in the current dataset.

Quantile	N	MeanRegressor	MeanOutcome
Q1	500	5.306	12.67
Q2	458	22.81	61.30
Q3	496	37.70	139.7
Q4	473	49.38	300.7

Quantile	N	MeanRegressor	MeanOutcome
Q5	451	66.92	657.9

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Next-year patents	Direct ties	Indirect ties	Structural holes	Log patent stock	Log assets	R&D intensity	Current-year patents	Current-year publications
Next-year patents	1	0.418	-0.349	0.228	0.476	0.436	0.113	1	1
Direct ties	0	1.000	-0.412	0.597	0.775	0.566	0.054	0	0
Indirect ties	0	-0.412	1.000	0.028	-0.236	-0.500	0.021	0	0
Structural holes	0	0.597	0.028	1.000	0.530	0.271	-0.022	0	0
Log patent stock	0	0.775	-0.236	0.530	1.000	0.536	0.158	0	0
Log assets	0	0.566	-0.500	0.271	0.536	1.000	-0.138	0	0

Compact correlation structure for the main variables.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A firm and year fixed-effects count model of next-year patent output on direct ties, indirect ties, their interaction, and structural holes. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Ahuja, 2000a; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The first set of replication decisions concerns scope. The unit of analysis is Firm-year, the sample frame is Public-firm year panel with rolling knowledge-link network measures and next-year patent output, and the empirical window is 1991 to 2014 in the local executable panel. These choices define what counts as a comparable observation in the local

reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Next-year patent count at the firm-year level, and the main explanatory design is Firm and year fixed-effects count model with direct ties, indirect ties, their interaction, and structural holes. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Patent stock, assets, and R&D intensity plus firm and year fixed effects.. The crucial exclusion or substitution rule is: The local rerun keeps only firms and years supported by the processed public-firm panel and the rolling citation-link network reconstruction.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

DecisionPoint	LocalSpecification
Research question	The source paper asks how direct ties, indirect ties, and structural holes in an interorganizational network shape subsequent innovation.
Unit of analysis	Firm-year

DecisionPoint	LocalSpecification
Sample frame	Public-firm year panel with rolling knowledge-link network measures and next-year patent output.
Time window	1991 to 2014 in the local executable panel.
Outcome definition	Next-year patent count at the firm-year level.
Main explanatory design	Firm and year fixed-effects count model with direct ties, indirect ties, their interaction, and structural holes.
Controls and fixed effects	Patent stock, assets, and R&D intensity plus firm and year fixed effects.
Exclusion or substitution logic	The local rerun keeps only firms and years supported by the processed public-firm panel and the rolling citation-link network reconstruction.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Firm-year inside the sample frame Public-firm year panel with rolling knowledge-link network measures and next-year patent output over 1991 to 2014 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about

the mechanism under study. Measurement decisions: the outcome is Next-year patent count at the firm-year level and the main explanatory design is Firm and year fixed-effects count model with direct ties, indirect ties, their interaction, and structural holes. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Patent stock, assets, and R&D intensity plus firm and year fixed effects. The main exclusion or substitution rule is The local rerun keeps only firms and years supported by the processed public-firm panel and the rolling citation-link network reconstruction. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$E[\text{NextPatents}_{it+1} \mid X_{it}] = \exp\left(\beta_1 \text{DirectTies}_{it} + \beta_2 \text{IndirectTies}_{it} + \beta_3 \text{DirectTies}_{it} \times \text{IndirectTies}_{it} + \beta_4 \text{Structurall} \right. \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is -0.0125 on Direct Ties (p = ¡ .001). The local citation-network analogue reverses the original alliance-network pattern in important ways: direct ties are negative, structural holes are positive, and indirect ties are weak once stock and scale controls are imposed. Log patent stock is estimated at 0.718 with p = ¡ .001. Direct ties is estimated at -0.0125 with p = ¡ .001. Log assets is estimated at 0.249 with p = ¡ .001 (Ahuja, 2000a; Hall et al., 2005). The coefficient table below reports the compact pattern most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Term	Estimate	Std. error	P-value
Direct ties	-0.0125	0.0033	0.0002
Indirect ties	-0.0008	0.0018	0.652
Structural holes	0.8257	0.3850	0.0320
Log patent stock	0.7182	0.0598	3.45e-33
Log assets	0.2495	0.0731	0.0006

Term	Estimate	Std. error	P-value
R&D intensity	3.989	1.234	0.0012
Direct ties x indirect ties	8.35e-05	5.35e-05	0.118

Key coefficient table used in the main text.

6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Log patent stock is estimated at 0.718 with a p-value of ¡ .001; Direct ties is estimated at -0.0125 with a p-value of ¡ .001; Log assets is estimated at 0.249 with a p-value of ¡ .001; R&D intensity is estimated at 3.989 with a p-value of 0.001 (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Log patent stock, which is estimated at 0.718 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Next-year patents has mean 228.2 and standard deviation 519.3 in the local panel, while the main regressor Direct ties has mean 35.89 and standard deviation 21.87 (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 12.67 to 657.9 while the mean regressor ranges from 5.306 to 66.92. This does not substitute for the model, but it shows how much raw variation the specification must explain (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local citation-network analogue reverses the original alliance-network pattern in important ways: direct ties are negative, structural holes are positive, and indirect ties are weak once stock and scale controls

are imposed. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within baseline, the leading coefficient family reads as follows: Log patent stock = 0.718 (p = ¡ .001); Direct ties = -0.0125 (p = ¡ .001); Log assets = 0.249 (p = ¡ .001); R&D intensity = 3.989 (p = 0.001); Structural holes = 0.826 (p = 0.032). The largest positive estimate in this specification is R&D intensity at 3.989, while the largest negative estimate is Direct ties at -0.0125. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Ahuja (2000) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than others, and a formal replication paper should show that uneven pattern directly in the main text (Ahuja, 2000a; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Log patent stock with estimate 0.718 and p = ¡ .001 (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993;

Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Baseline specification contributes its own reading to the replication. Log patent stock is estimated at 0.718 with p = ¡ .001 and is classified as a strong signal; Direct ties is estimated at -0.0125 with p = ¡ .001 and is classified as a strong signal; Log assets is estimated at 0.249 with p = ¡ .001 and is classified as a strong signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Ahuja (2000) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

Term	Model	Estimate	PValue	Signal	Reading
Direct ties	Baseline	-0.0125	¡ .001	Strong	Negative estimate with p = ¡ .001.
Indirect ties	Baseline	-0.0008	0.65	Weak	Negative estimate with p = 0.65.
Structural holes	Baseline	0.8257	0.032	Moderate	Positive estimate with p = 0.032.
Log patent stock	Baseline	0.7182	¡ .001	Strong	Positive estimate with p = ¡ .001.
Log assets	Baseline	0.2495	¡ .001	Strong	Positive estimate with p = ¡ .001.
R&D intensity	Baseline	3.989	0.001	Strong	Positive estimate with p = 0.001.

Term	Model	Estimate	PValue	Signal	Reading
Direct ties x indirect ties	Baseline	8.35e-05	0.12	Weak	Positive estimate with p = 0.12.

Model-by-model interpretation table drawn from the current coefficient exports.

6.5 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005). For network_innovation, the current fit export reports N = 2,276, pseudo R-squared = 0.954 (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Ahuja (2000) is being carried out on a clearly described estimating sample with observable fit statistics (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005).

model	n	logLik	AIC	BIC	pseudo_r2
network_innovation	2,276	-25,666	51,700	52,754	0.954

Model fit statistics discussed in the main results section.

Term	Estimate	Std. error	Statistic	P-value
Direct ties	-0.0125	0.0033	-3.745	0.0002
Indirect ties	-0.0008	0.0018	-0.451	0.652
Structural holes	0.8257	0.3850	2.145	0.0320
Log patent stock	0.7182	0.0598	12.00	3.45e-33
Log assets	0.2495	0.0731	3.415	0.0006
R&D intensity	3.989	1.234	3.233	0.0012
Direct ties x indirect ties	8.35e-05	5.35e-05	1.562	0.118

Detailed coefficient table from the current local estimation output.

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (Ahuja, 2000a). For Ahuja (2000), the local headline result is Direct Ties = -0.0125 with p = ¡ .001. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper reports positive effects of direct and indirect ties on innovation, with structural holes exerting a negative effect in the interfirm collaboration network.	Direct Ties = -0.0125 with p = ¡ .001.	The local analogue does not confirm the original positive direct-ties logic. In the citation-based network, more direct ties are associated with lower next-year patenting, whereas structural holes are positive. That contrast suggests the alliance-network mechanism does not transport mechanically into a citation-network setting.

Aspect	OriginalPaper	LocalDataset	Reading
Measurement and data construction	The published network is a collaboration network among firms rather than a citation-based knowledge-link network.	The local analogue reconstructs a rolling firm-to-firm knowledge-link network from backward patent ties among public firms and then maps direct ties, indirect ties, and structural holes into a firm-year innovation panel.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.
Sample and unit of analysis	The published sample is a longitudinal study of firms in the international chemicals industry.	Public-firm year panel with rolling knowledge-link network measures and next-year patent output. Local sample = 2,378 observations across 177 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published network is a collaboration network among firms rather than a citation-based knowledge-link network By contrast, the local paper relies on the following executable design: The local analogue reconstructs a rolling firm-to-firm knowledge-link network from backward patent ties among public firms and then maps direct ties, indirect ties, and structural holes into a firm-

year innovation panel These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published sample is a longitudinal study of firms in the international chemicals industry The local sample instead works from Public-firm year panel with rolling knowledge-link network measures and next-year patent output and yields 2,378 observations across 177 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The local analogue does not confirm the original positive direct-ties logic. In the citation-based network, more direct ties are associated with lower next-year patenting, whereas structural holes are positive. That contrast suggests the alliance-network mechanism does not transport mechanically into a citation-network setting. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The local analogue does not confirm the original positive direct-ties logic. In the citation-based network, more direct ties are associated with lower next-year patenting, whereas structural holes are positive. That contrast suggests the alliance-network mechanism does not transport mechanically into a citation-network setting. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Ahuja, 2000a; March,

1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

The local analogue does not confirm the original positive direct-ties logic. In the citation-based network, more direct ties are associated with lower next-year patenting, whereas structural holes are positive. That contrast suggests the alliance-network mechanism does not transport mechanically into a citation-network setting (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Ahuja, 2000a; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000b; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The source paper is based on collaboration networks in the chemicals industry (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. The local rerun uses a knowledge-link network built from patent citations among public firms rather than alliances (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The exercise therefore evaluates the network mechanism on a different but transparent interfirm relation (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a transparent network translation that makes the tie-definition shift explicit and shows how the innovation coefficients change when knowledge links replace collaborative alliances (Ahuja, 2000a; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Ahuja, 2000a,b; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	2,378	rows
Unique Firms	177	unique_firms
Start Year	1991	year_min
End Year	2014	year_max
Mean Direct Ties	35.89	mean_direct_ties
Mean Indirect Ties	52.56	mean_indirect_ties
Mean Structural Holes	0.256	mean_structural_holes
Mean Next Patents	228	mean_next_patents

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Next-year patents	2,378	228.2	519.3	20.00	78.50	226.4	0	7662
Direct ties	2,378	35.89	21.87	18.00	37.00	51.00	0	96
Indirect ties	2,378	52.56	26.60	31.00	54.00	75.00	0	105
Structural holes	2,378	0.256	0.156	0.158	0.255	0.354	0	1
Log patent stock	2,378	5.803	2.097	4.833	6.254	7.236	0	10
Log assets	2,378	9.388	1.341	8.480	9.310	10.26	4	14
R&D intensity	2,378	0.0417	0.0403	0.0134	0.0281	0.0594	0	1
Current-year patents	2,378	218.2	489.0	20.00	77.00	217.1	0	7661
Current-year publications	2,378	146.6	301.6	3.500	23.00	154.0	0	2459

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Next-year patents	Direct ties	Indirect ties	Structural holes	Log patent stock	Log assets	R&D intensity	Current-year patents	Current-year publications
Next-year patents	1	0.418	-0.349	0.228	0.476	0.436	0.113	1	1
Direct ties	0	1.000	-0.412	0.597	0.775	0.566	0.054	0	0
Indirect ties	0	-0.412	1.000	0.028	-0.236	-0.500	0.021	0	0
Structural holes	0	0.597	0.028	1.000	0.530	0.271	-0.022	0	0
Log patent stock	0	0.775	-0.236	0.530	1.000	0.536	0.158	0	0
Log assets	0	0.566	-0.500	0.271	0.536	1.000	-0.138	0	0
R&D intensity	0	0.054	0.021	-0.022	0.158	-0.138	1.000	0	0
Current-year patents	1	0.431	-0.355	0.238	0.486	0.439	0.110	1	1
Current-year publications	1	0.355	-0.232	0.267	0.430	0.461	0.270	1	1

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	FirmYears	MeanDirectTies	MeanIndirectTies	MeanStructuralHoles	MeanNextPatents
1991	132	23.24	68.59	0.377	118
1992	130	28.17	68.15	0.352	122
1993	126	31.56	69.62	0.321	130
1994	123	34.60	66.11	0.292	129
1995	122	37.31	69.41	0.287	140
1996	123	37.32	72.59	0.287	130
1997	120	38.48	71.18	0.283	170
1998	113	38.71	66.27	0.282	183
1999	106	38.83	57.91	0.270	195
2000	102	35.86	47.63	0.234	209
2001	100	38.10	48.88	0.233	219
2002	98	38.04	53.18	0.238	226
2003	100	39.50	51.58	0.235	216
2004	98	39.08	49.20	0.241	195
2005	93	40.26	45.68	0.235	250
2006	89	40.40	39.73	0.229	232
2007	83	40.12	33.93	0.205	258
2008	78	37.77	31.51	0.198	292
2009	75	35.28	30.99	0.186	387
2010	75	35.07	30.45	0.181	374
2011	72	34.72	30.69	0.179	425
2012	70	34.91	25.17	0.179	463
2013	75	35.12	26.45	0.175	468
2014	75	35.76	27.33	0.178	463

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	500	5.306	12.67
Q2	458	22.81	61.30
Q3	496	37.70	139.7
Q4	473	49.38	300.7
Q5	451	66.92	657.9

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value
direct_ties	-0.0125	0.0033	-3.745	¡ .001
indirect_ties	-0.0008	0.0018	-0.451	0.65
structural_holes	0.8257	0.3850	2.145	0.032
log_pat_stock	0.7182	0.0598	12.00	¡ .001
log_assets	0.2495	0.0731	3.415	¡ .001
rd_intensity	3.989	1.234	3.233	0.001
direct_ties:indirect_ties	8.35e-05	5.35e-05	1.562	0.12

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Next-year patents	0	0.00%
Direct ties	0	0.00%
Indirect ties	0	0.00%
Structural holes	0	0.00%
Log patent stock	0	0.00%
Log assets	0	0.00%
R&D intensity	0	0.00%
Current-year patents	0	0.00%
Current-year publications	0	0.00%

Missingness audit for the variables summarized in the manuscript.

Entity	Label	Observations	FirstPeriod	LastPeriod
10499	TEXAS INSTRUMENTS INC	24	1991	2014
10519	TEXTRON INC	24	1991	2014
1078	ABBOTT LABORATORIES	24	1991	2014
10983	RTX CORP	24	1991	2014
11465	WHIRLPOOL CORP	24	1991	2014
12142	ORACLE CORP	24	1991	2014
1300	HONEYWELL INTERNATIONAL INC	24	1991	2014
1356	ALCOA INC	24	1991	2014
1690	APPLE INC	24	1991	2014
1722	ARCHER-DANIELS-MIDLAND CO	24	1991	2014
2285	BOEING CO	24	1991	2014

Entity	Label	Observations	FirstPeriod	LastPeriod
2403	BRISTOL-MYERS SQUIBB CO	24	1991	2014
2663	CAMPBELL'S CO (THE)	24	1991	2014
2817	CATERPILLAR INC	24	1991	2014
2991	CHEVRON CORP	24	1991	2014
3170	COLGATE-PALMOLIVE CO	24	1991	2014
3310	CA INC	24	1991	2014
3532	CORNING INC	24	1991	2014
3835	DEERE & CO	24	1991	2014
4058	DOVER CORP	24	1991	2014

Top entities by local observation count.

References

Ahuja, G. (2000a). Collaboration networks, structural holes, and innovation: A longitudinal study. *Administrative Science Quarterly*, 45(3):425–455.

Ahuja, G. (2000b). Collaboration networks, structural holes, and innovation: A longitudinal study. *Administrative Science Quarterly*, 45(3):425–455.

Ahuja, G. and Katila, R. (2001). Technological acquisitions and the innovation performance of acquiring firms: A longitudinal study. *Strategic Management Journal*, 22(3):197–220.

Ahuja, G. and Lampert, C. M. (2001). Entrepreneurship in the large corporation: A longitudinal study of how established firms create breakthrough inventions. *Strategic Management Journal*, 22(6-7):521–543.

Almeida, P. and Kogut, B. (1999). Localization of knowledge and the mobility of engineers in regional networks. *Management Science*, 45(7):905–917.

Anderson, P. and Tushman, M. L. (1986). Technological discontinuities and organizational environments. *Administrative Science Quarterly*, 31(3):439–465.

Benner, M. J. and Tushman, M. L. (2003). Exploitation, exploration, and process management: The productivity dilemma revisited. *Academy of Management Review*, 28(2):238–256.

Cohen, W. M. and Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1):128–152.

Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, 47(1):117–132.

Fleming, L., Mingo, S., and Chen, D. (2007). Collaborative brokerage, generative creativity, and creative success. *Administrative Science Quarterly*, 52(3):443–475.

Hall, B. H., Jaffe, A., and Trajtenberg, M. (2005). Market value and patent citations. *RAND Journal of Economics*, 36(1):16–38.

Hargadon, A. and Sutton, R. I. (1997). Technology brokering and innovation in a product development firm. *Administrative Science Quarterly*, 42(4):716–749.

Henderson, R. and Cockburn, I. (1994). Measuring competence? exploring firm effects in pharmaceutical research. *RAND Journal of Economics*, 25(1):63–84.

Henderson, R. M. and Clark, K. B. (1990). Architectural innovation: The reconfiguration of existing product technologies and the failure of established firms. *Administrative Science Quarterly*, 35(1):9–30.

Jaffe, A. B. (1986). Technological opportunity and spillovers of r&d: Evidence from firms' patents, profits, and market value. *American Economic Review*, 76(5):984–1001.

Jaffe, A. B., Trajtenberg, M., and Henderson, R. (1993). Geographic localization of knowledge spillovers as evidenced by patent citations. *Quarterly Journal of Economics*, 108(3):577–598.

Katila, R. and Ahuja, G. (2002). Something old, something new: A longitudinal study of search behavior and new product introduction. *Academy of Management Journal*, 45(6):1183–1194.

Kogut, B. and Zander, U. (1992). Knowledge of the firm, combinative capabilities, and the replication of technology. *Organization Science*, 3(3):383–397.

Kogut, B. and Zander, U. (1996). What firms do? coordination, identity, and learning. *Organization Science*, 7(5):502–518.

Lavie, D., Kang, J., and Rosenkopf, L. (2011). Balance within and across domains: The performance implications of exploration and exploitation in alliances. *Organization Science*, 22(6):1517–1538.

Levinthal, D. A. and March, J. G. (1993). The myopia of learning. *Strategic Management Journal*, 14(S2):95–112.

Levitt, B. and March, J. G. (1988). Organizational learning. *Annual Review of Sociology*, 14:319–338.

March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87.

Rivkin, J. W. and Siggelkow, N. (2003). Balancing search and stability: Interdependencies among elements of organizational design. *Management Science*, 49(3):290–311.

Rosenkopf, L. and Almeida, P. (2003). Overcoming local search through alliances and mobility. *Management Science*, 49(6):751–766.

Rosenkopf, L. and Nerkar, A. (2001). Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4):287–306.

Schilling, M. A. and Phelps, C. C. (2007). Interfirm collaboration networks and knowledge creation: The impact of large-scale network structure on firm innovation. *Management Science*, 53(7):1113–1126.

Stuart, T. E. (2000). Interorganizational alliances and the performance of firms: A study of growth and innovation rates in a high-technology industry. *Strategic Management Journal*, 21(8):791–811.

Stuart, T. E. and Podolny, J. M. (1996). Local search and the evolution of technological capabilities. *Strategic Management Journal*, 17(S1):21–38.

Teece, D. J., Pisano, G., and Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal*, 18(7):509–533.

Trajtenberg, M. (1990). A penny for your quotes: Patent citations and the value of innovations. *RAND Journal of Economics*, 21(1):172–187.

Tripsas, M. and Gavetti, G. (2000). Capabilities, cognition, and inertia: Evidence from digital imaging. *Strategic Management Journal*, 21(10-11):1147–1161.